TEXORA · LECTURE SHEET

Jacquard Shedding

The shedding system that controls warp ends individually instead of in shafts: a perforated card decides, for every pick, which of several hundred hooks the rising griffe shall lift — making damask, brocade and tapestry possible.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 6 · LEVEL 2 TERM 1

1. Why the jacquard exists — beyond the dobby

Both the tappet and the dobby move warp ends **in groups**: every end on one shaft rises and falls together, so no two ends on the same shaft can ever behave differently. A figured design breaks that limit at once: the repeat may be hundreds of ends wide, with neighbouring ends rising on completely different pick sequences — and giving every end its own shaft would need hundreds of shafts, which no loom can carry.

The **jacquard machine**, mounted above the loom and named after Joseph-Marie Jacquard, solves the problem by abandoning heald shafts for the figure: there are **no heald shafts** in a jacquard harness. Instead, **each hook controls a group of warp ends through cords** — with a straight tie, one end in each repeat — so ends are controlled **individually within the repeat**. The pattern is carried by **perforated pasteboard cards**, one per pick, whose holes decide which hooks are lifted. Change the cards, and the machine weaves a wholly different figure.

2. Components of the jacquard machine

- **Hooks.** Vertical wires in rows, each with a bent beak at top and a loop at bottom for the cords. Their number is the machine's capacity; a lifted hook raises every end tied to it.
- **Needles.** Horizontal wires, one per hook, each passing through a loop in its hook, so pushing the needle **deflects its hook sideways**. Needles read the card; hooks lift.
- **Spring box.** Houses a spring for every needle. The springs press the needles toward the cylinder, return them after each push, and so keep every hook over the griffe until a card displaces it.
- **Griffe (lifting block).** A frame of blades set edge-up, one under each hook row, rising and falling once per pick. Hooks over the blades are lifted; deflected hooks are missed.
- **Card cylinder.** A square prism that presses the card against the needle ends, swings away, turns a quarter-turn and presents the next card, timed to the lifting cycle.

- **Perforated cards.** Punched pasteboard cards laced into an endless chain over the cylinder; one card per pick. Hole: needle not pushed. Blank: needle pushed back.
- **Neck cords.** Short cords from the hook loops down through holes in the **comber board (harness board)**, which keeps them in order and spacing.
- **Harness cords.** The long cords continuing from the neck cords toward the warp, each carrying a **heald (mail)** — a small strip with an eye through which one warp end is drawn.
- **Lingoes.** Slender cast-iron weights, one below each heald. The lingo tensions the cord, pulls the end **down again** after lifting (the jacquard lifts positively but returns by weight) and holds unselected ends on the bottom line. Classical jacquard shedding is thus **single-lift with weight return**.

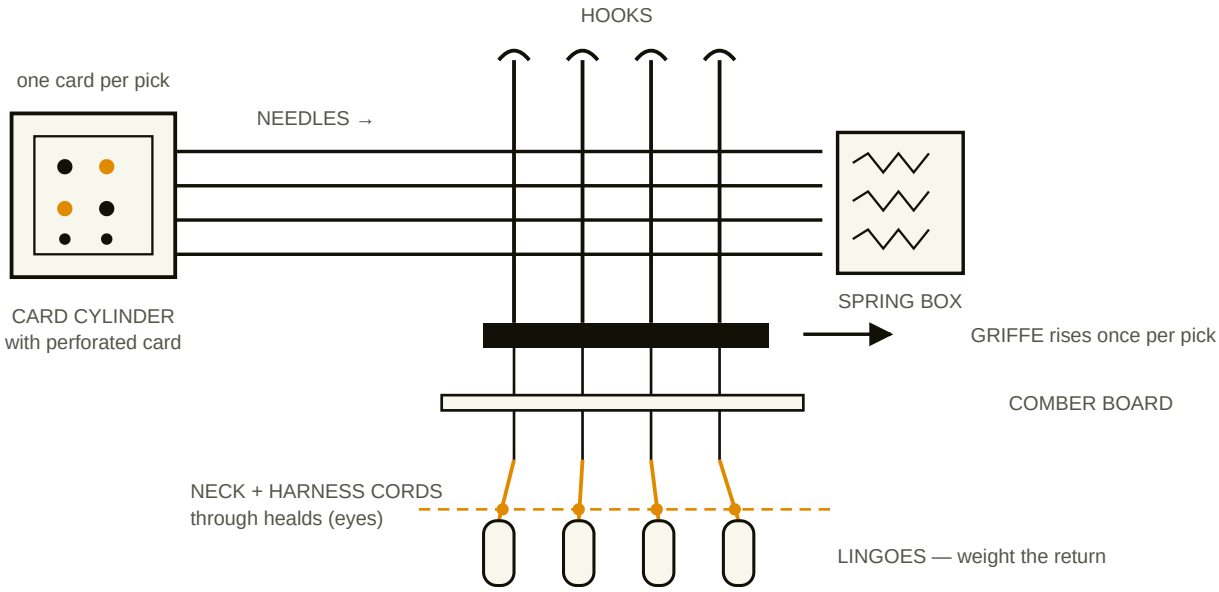


Figure 1 — Jacquard head and harness, schematic (a few hooks only; a real machine has hundreds). Unpushed needles leave their hooks over the griffe, which lifts them and their ends; lingoes hold the rest on the bottom line.

3. The working cycle — hole lifts, blank leaves down

The whole selection logic of the classical jacquard rests on one contrast, and it must be stated exactly:

- **Hole present** → the needle facing the hole **passes into it** and is **not pushed back**; its hook **stays over a griffe blade**, and the rising griffe **lifts it** — and every end tied to it. **A hole means lift.**
- **Blank (no hole)** → the solid card **pushes the needle back against its spring**, deflecting its hook **off the blade**; the rising griffe misses it, and the lingoes hold its ends on the bottom line. **A blank means no lift.**

The full cycle, once per pick, runs as follows:

1. **Presentation.** The cylinder swings in and presses this pick's card firmly against the needle ends — needles enter the holes where holes exist; blanks push their needles back. Selection is now complete, stored mechanically in the positions of the hooks.
2. **Cylinder withdrawal.** The cylinder swings clear of the needles and turns a quarter-turn, bringing the next card into position.
3. **Lift.** The griffe rises, carrying up every hook still over a blade; through neck cord, harness cord and heald, each hook draws its ends to the top line. Deflected hooks stay down. The shed is open.
4. **Weft insertion and beat-up.** The pick is inserted and beaten up in the normal way.
5. **Return.** The griffe falls; the hooks descend with it and the lingoes draw their ends down by weight, restoring the bottom line. The next card is presented, and the cycle repeats.

Note the timing demand: **selection must finish before the griffe rises**, and the cylinder must clear the needles before it falls — otherwise hooks are dragged while loaded, needles bend, and single-pick faults are woven permanently into the cloth.

4. Capacity — hook counts and what they buy

A jacquard machine is described by its **hook count**. Common classical sizes are **400, 600, 900 and 1200 hooks**, with larger machines of 1200+ hooks for very wide repeats; a "600s" machine has about 600 usable hooks. Two consequences follow:

- **Design repeat width.** With a straight tie at one end per hook, hooks equal ends per repeat (Section 6): a 600-hook machine weaves a figure 600 ends wide before the pattern repeats across the cloth.
- **Design repeat length.** The pick repeat is the number of cards in the lace — one card per pick — so long repeats cost cards, not machine size. Laces of a thousand cards and more are routine.

Electronic jacquards replace the card chain with electronically controlled selection: each hook is selected by a signal from the pattern file, with no cards to cut, lace or store. Hook counts run into several thousand on wide machines, and the hole-versus-blank logic of Section 3 survives unchanged in software — a stored "1" lifts, exactly as a punched hole did.

5. The harness — tying up the machine to the warp

Between the hooks and the warp stands the **harness**: neck cords through the comber board, harness cords down to the healds, a lingo below each heald. How the cords are distributed across the warp is the **tie-up**, and it fixes how the design repeat lies in the cloth.

5.1 Straight tie

In a **straight tie** the hooks are tied in order across the warp: hook 1 serves the first end of the repeat, hook 2 the second, and so on; when all hooks are used once, the sequence restarts for the next repeat. The design repeats **without mirroring**, exactly as drawn. Straight tie is the standard for damasks and most figured work, and is the tie assumed in Section 6.

5.2 Pointed (reversing) tie

In a **pointed tie** the hook sequence runs across the repeat and then **reverses**, like a mirror: 1, 2, 3 … to the last hook and back down again. Each hook (except the two turning hooks) is used twice per cycle, so the machine covers roughly **double the repeat width** — but the design is **mirrored at every point**. Pointed ties suit symmetrical motifs (diamonds, centred sprays) and are impossible for lettering or any unsymmetrical figure, which would read backwards on alternate repeats.

5.3 Other ties in outline

Border ties reserve separate hook groups for the selvedge borders and the body. **Split ties** divide the hooks between interleaved repeats or figure and ground. In every case: count the ends one hook controls per repeat, and repeats across follow by division. A single cord tied to the wrong hook displaces its ends for the whole piece, so ties are checked end by end before weaving.

6. Design repeat and card cutting

A jacquard design is prepared on squared **design paper**, one square per end-and-pick crossing, figure painted in and weave structure marked. A card-cutting machine then punches the cards row by row — in each card exactly one pick's holes: hole where an end must rise, blank where it stays down (Section 3). The cards are numbered and laced in order into a chain (a **lace**); one wrong hole repeats its error in every repeat, so cards are proofed against the design over one full repeat.

Two pieces of arithmetic govern every jacquard job, both simple divisions once the rule "one card per pick, one hook per end per repeat" is held in mind:

WORKED EXAMPLE — CARDS NEEDED FOR A DESIGN REPEAT

A damask design repeats every 800 picks. Cards are cut for one full repeat lace.
Step 1 — rule: one card is read for each pick, so cards and picks are in one-to-one correspondence.
Step 2 — cards per repeat lace = 800 picks × 1 card per pick = **800 cards per repeat lace**.
Step 3 — check: the lace joins end to end, so after card 800 card 1 returns and the design restarts exactly; any other count would drift the figure by a pick at the joint.

WORKED EXAMPLE — ENDS CONTROLLED AND REPEATS ACROSS THE WIDTH

A 600-hook jacquard is tied straight, with one end per hook in each repeat. The cloth contains 4800 warp ends.
Step 1 — ends per repeat = hooks in use × ends per hook = 600 × 1 = **600 ends per repeat**.
Step 2 — repeats across the width = total ends ÷ ends per repeat = 4800 ÷ 600 = **8 repeats across the width**.
Step 3 — check: 8 × 600 = 4800 ends, exactly the warp — the figure repeats eight times with no part-repeat at either selvedge.

Cards per repeat = picks in the design repeat · Repeats across = total ends ÷ (hooks × ends per hook per repeat)

Making the figure *wider* needs a bigger machine (more hooks); making it *longer* needs only more cards.

7. Setting, care and common faults

- **Timing.** The griffe must rise only after the card is fully read, and the cylinder must swing clear before it falls; mistiming bends needles.
- **Hook and needle condition.** A bent needle holds its hook deflected (its ends never lift); a worn beak slips off the blade. Trace faulty ends by lifting the harness by hand, hook by hook.
- **Card care.** Keep cards dry, flat and in order; a swollen or torn card misreads, and a lace assembled out of order scrambles the design.
- **Harness condition, levelling and cleanliness.** Frayed cords, cracked healds and unequal lingoes give slack ends and a ragged bottom line; keep lingo weights uniform and board holes free of fluff. At rest every end must lie on one bottom line; re-check level and shed height after any re-tieing, and brush down the needle board and spring box daily.

8. Jacquard versus dobby versus tappet — comparison

FEATURE	TAPPET	DOBBY	JACQUARD
Pattern unit controlled	Heald shaft (group of ends)	Heald shaft (group of ends)	Individual end within the repeat
Pattern stored as	Cam profiles	Pegs in lags, tape holes, or a file	Punched cards, one per pick — or a file
Heald shafts	About 8 (rarely 10)	Commonly 12–24	None for the figure — a cord harness instead

Ends in one design repeat	Limited by shafts and draft	Limited by shafts and draft	Equal to hooks in use — 400, 600, 900, 1200+
Pick repeat	Short, fixed by cams	Long — one lag per pick	Long — one card per pick; 1000+ card laces routine
Typical cloths	Plain weave, simple twills	Checks, multi-shaft twills, geometrics	Damask, brocade, tapestry, figured fabrics
Return of lifted ends	Positive (cam) or spring	Springs (negative) or driven (positive)	By lingo weight (classical machines)
Design change	New cams — slow, costly	New chain or file — quick	New cards (costly to cut) or a file (instant)
Cost, size and speed	Cheapest, smallest, fast	Intermediate	Costliest and bulkiest; slowest of the three

Every large figured fabric is still woven on Jacquard's principle, in pasteboard or in a pattern file.

Glossary

Jacquard machine — a shedding machine above the loom selecting hooks individually from cards (or electronically). **Hook** — a vertical wire whose lifting raises all ends tied to it. **Needle** — a horizontal wire reading the card, deflecting its hook on or off the griffe. **Spring box** — housing of the needle springs. **Griffe** — the rising blade frame lifting hooks left over its blades. **Card cylinder** — the prism pressing each card against the needles. **Card** — one perforated sheet per pick; hole = lifted, blank = not lifted. **Lace** — the card chain in pick order. **Neck cord** — the cord from a hook through the comber board. **Harness** — the cord system connecting hooks to ends. **Heald (mail)** — the eye through which one end is drawn. **Lingo** — the weight below each heald, returning the end. **Comber board** — the drilled board spacing the neck cords. **Straight tie** — hooks serve successive ends in order. **Pointed tie** — the hook sequence reverses at each repeat, mirroring the design.

Key points

• The jacquard controls ends individually within the repeat; no heald shafts are used for the figure.
• Hole → needle passes through → hook stays over the griffe → lifted. Blank → needle pushed → hook deflected → not lifted.
• One card per pick; the pick repeat is the number of cards in the lace.
• Capacities: 400, 600, 900, 1200+ hooks; electronic machines run to several thousand.
• Ends per repeat = hooks × ends per hook; repeats across = total ends ÷ ends per repeat.

• Straight ties repeat the design as drawn; pointed ties mirror it and roughly double the width.
• Lift is by the griffe; return is by lingo weight; selection must finish before the griffe moves.

Practice

1. How many cards must be cut for a lace holding three repeats of a 1200-pick design? Show the steps.
2. A 900-hook jacquard, straight tie, one end per hook, weaves a cloth of 5400 ends. How many repeats lie across the width? Show the steps.
3. A card has a hole facing needle 37 and a blank facing needle 38. For each, state what happens to the needle, its hook and its ends when the griffe rises — and why.
4. A customer wants a wide symmetrical diamond motif and, separately, woven lettering. Which tie for each, and why?
5. Several ends controlled by one hook never rise, while all other ends weave correctly. Give the two most likely machine-side causes and how you would confirm each.
6. Why is lengthening a jacquard design cheap but widening it expensive? Name what each direction consumes.

Answer key

1. 3 × 1200 = **3600 cards** (one card per pick, so one repeat needs 1200).
2. Ends per repeat = 900 × 1 = 900 ends. Repeats across = 5400 ÷ 900 = **6 repeats**. Check: 6 × 900 = 5400 ends.
3. Needle 37 (hole): not pushed back; its hook stays over the blade and is lifted, raising its ends. Needle 38 (blank): pushed back; its hook is deflected off the blade and its ends stay down.
4. Diamond: pointed tie — its mirroring suits a symmetrical motif and roughly doubles the width. Lettering: straight tie — a pointed tie would mirror alternate repeats and the letters would read backwards.
5. (i) The needle is bent, holding the hook permanently deflected — confirm by inspecting it and testing whether the hook can reach the blade by hand. (ii) The hook is bent or its beak damaged — confirm by raising the griffe slowly by hand and watching that hook stand aside. (Check the neck cord too.)
6. Length consumes **cards**: one cheap extra card per pick. Width consumes **hooks**: each extra end needs a hook with needle, spring, cords, heald and lingo.